

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 3
 - Set basics
 - Set algebra

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- $\{n \in \mathbb{Z} : n \text{ is prime}\}$

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- even relations and functions are sets!

Challenge!

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Prove that for ordered pairs (a, b) and (c, d) that

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Case I: $a = c$ and $\{a, b\} = \{c, d\}$

Case II: $a = \{c, d\}$ and $\{a, b\} = c$

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Therefore $b = c$ or $b = d$.

If $b = d$, we're done!

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Then $a = c$ implies $a = b$.

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It follows that $d = c = b = a$.

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This would imply that $c \in a$ and $a \in c$.

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This can be shown to contradict the ZF Axioms of Set Theory.

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This can be shown to contradict the ZF Axioms of Set Theory.
Specifically the regularity axiom for the set $\{a, c\}$...

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NOTATION:

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Challenge

- index set $I = (1, 2)$
- family of sets $\{A_i : i \in I\}$
- $A_i = [0, i]$

Problem

Determine $\bigcup_{i \in I} A_i$.

Challenge

- index set $I = \mathbb{Z}_+$
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Problem

Determine $\bigcap_{i \in I} A_i$.

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Prove that De Morgan's Laws apply to families of sets too, ie.

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- $\leq = \{(x, y) : y - x \in [0, \infty)\}$ is reflexive and transitive but not symmetric on $\mathbb{R}$

Challenge!

Problem

Give an example of a relation on $\mathbb{R}$ which is symmetric and transitive but not reflexive.

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NOTATION: $f : A \rightarrow B$ means f is a function from A to B

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The set

$$\text{img}(f) = \{f(a) : a \in A\}$$

is called the **range** or **image** of f

Challenge!

Problem

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Thus the only function which is an equivalence relation is the identity function

$$f(x) = x.$$